Calibration using autocorrelated visibilities with MWA

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1 Introduction

This memorandum explains the role of autocorrelated visibilities in radio interferometry and evaluates their use as a scaling mechanism during calibration of interferometric data, particularly in the context of Murchison Widefield Array (MWA) processing workflows.

In radio interferometry, observed data consist of visibilities, which are complex correlations between antenna signals:

- Cross-correlations: between different antennas ij

- Autocorrelations: correlation of a signal with itself $i = j$

While cross-correlations encode spatial information about the sky, autocorrelations primarily measure total received power, including:

1. sky brightness temperature
2. receiver noise
3. instrumental systematics

Autocorrelations are often excluded from imaging pipelines due to their susceptibility to noise and systematics. However, they provide valuable information for amplitude scaling and calibration.

2 Autocorrelated Visibilities

An autocorrelation visibility can be expressed as:

$$V_{ii}(\nu) \approx |g_i(\nu)|^2 \cdot (T_{sky} + T_{rec}) \tag{1}$$

where $g_i(\nu)$ is the complex gain of antenna i, T_{sky} is the sky temperature and T_{rec} refers to instrumental noise

3 Scaling Autocorrelations

Autocorrelations can be used to normalise or scale gain solutions derived from cross-correlations. This is particularly useful when:

Absolute flux calibration is uncertain Gain amplitudes drift over time or frequency Bandpass solutions require stabilisation Common approach: Compute autocorrelation power per antenna Derive a scaling factor:

$$S_i(\nu) \propto \sqrt{V_{ii}(\nu)} \tag{2}$$

Apply scaling to cross-correlation gains:

$$g_i^{scaled}(\nu) = \frac{g_i(\nu)}{S_i(\nu)} \tag{3}$$

This ensures consistency between measured total power (autocorrelations) and interferometric amplitudes (cross-correlation)

4 Application in MWA Calibration Pipelines

5 Conclusion

Autocorrelated visibilities provide a valuable but imperfect tool for scaling calibration solutions in radio interferometry. When used carefully, they enhance amplitude stability and offer independent constraints on system behaviour. However, due to their sensitivity to noise and systematics, they should be used in conjunction with cross-correlation-based calibration methods rather than as a standalone solution.